

K2-30b

R-2425

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-30_201125 · created 2026-08-22T00:03:14+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459178.6076 +/- 0.0084 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.111 +/- 0.043
Transit depth (Rp/Rs)^2	1.23 +/- 0.95 [%]
Orbital Inclination (inc)	86.5 +/- 2.24
Airmass coefficient 1 (a1)	0.999 +/- 0.025
Airmass coefficient 2 (a2)	0.0007 +/- 0.0079
Scatter in the residuals of the lightcurve fit is	2.48 %
Best Comparison Star	#3 - [134, 195]
Optimal Aperture	3.17
Optimal Annulus	16.53
Transit Duration (day)	0.065 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.111 ± 0.043 vs 0.1266 (deviation 0.36σ)
Duration fit vs published	0.065 d vs 0.0971 d (deviation 0.94σ)
Mid-time O–C	-19.9 min
Depth SNR	2.6
Residual scatter	2.48 %

Light curve

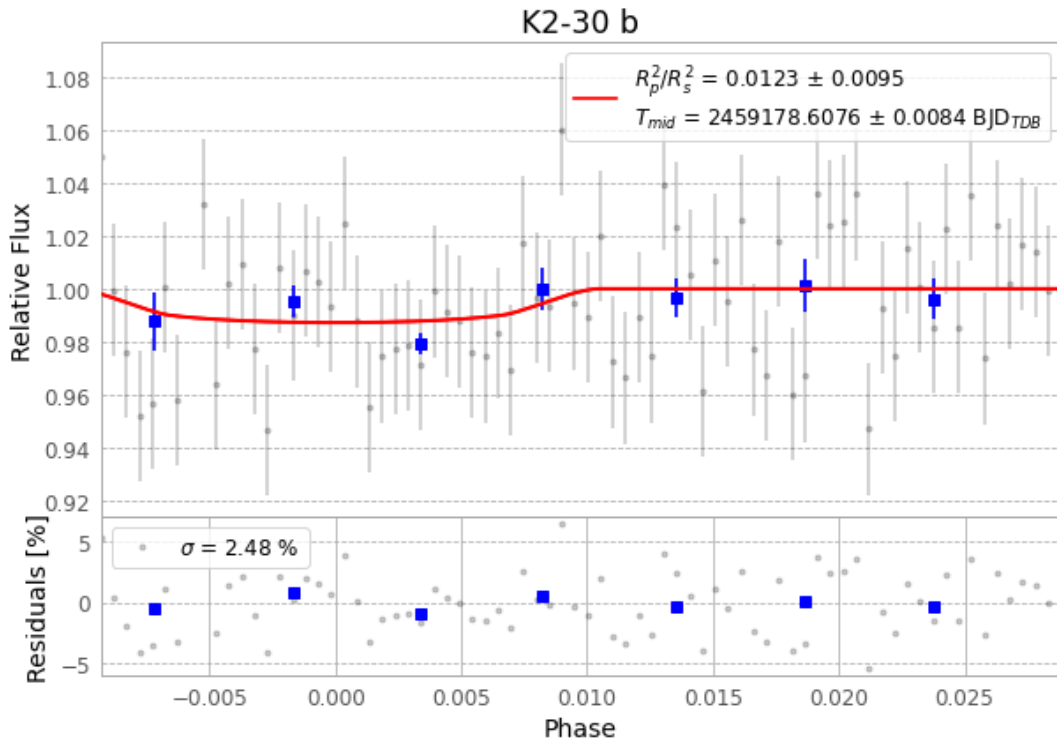


Figure 1 — FinalLightCurve_K2-30 b_2020-11-24.png (SHA-256 c18c81ec2af84ae8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [495, 222], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 2c7d067231dea9aa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

76 FITS frames (50 MB), K2-30201125013012.FITS → K2-30201125051513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-30-201125.log (7ba0fa98e505...), FinalLightCurve_K2-30 b_2020-11-24.png (c18c81ec2af8...), FinalLightCurve_K2-30 b_2020-11-24.csv (5f68a6d03308...)

Compute resources

Wall time 97.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-22 00:03Z.